



Facile preparation of structured zwitterionic polymer substrate via sub-surface initiated atom transfer radical polymerization and its synergistic marine antifouling investigation

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ABSTRACT

In this paper, using one-step method of sub-surface initiated atom transfer radical polymerization (SSI-ATRP), the novel polymer brushes accompanied with structured surface were prepared on the resin base. Compared with the traditional surface initiated atom transfer radical polymerization (SI-ATRP) method, either surface chemical composition or topography was obtained through SSI-ATRP synchronously. Besides, we found that the structured polymers brush interface had better stability and long-term synergistic antifouling effect under seawater medium through soaking experiment and ocean field assay for long time. While, the bio-fouling assay results indicated that their excellent antifouling performances would be hindered by salt responsiveness of 3-sulfopropyl methacrylate potassium salt (SPMA) polymer brushes. Because it could be screened by high salt (e.g. Ca^{2+} or Mg^{2+}) concentration, which decreased the hydration effect and further weakened the antifouling effect. The zwitterionic polymer (sulfobetaine methacrylate, SBMA), however, still showed excellent antifouling performances under high salt concentration since it produced stabilized ionic bonds with water molecules than those created from other hydrophilic materials. Given this, the structured zwitterionic and anionic copolymer brushes were fabricated as well, which realized the ideal stability and long-term antifouling effect under marine environment, even high salt concentration medium. This research is of great value to marine antifouling in the long run and some other relevant fields.

1. Introduction

Marine bio-fouling, which is subject to the undesirable settlement of marine organisms (bacteria, algae, mollusks), protein and bacteria, causes the damages liking increase of fuel consumption, corrosion of hull, obstruction of underwater pipes and invasion of alien species [1]. With the development of marine transportation and aquaculture, marine bio-fouling has been one of the most critical global challenges humanity faced, costing billions of dollars per year in transportation [2,3]. In terms of present situation, some vital marine antifouling (AF) strategies have been presented and could reduce the damages caused by marine bio-fouling.

The traditional techniques first offered were pitch, tar, wax, heavy metals (lead), or toxic (arsenic-based) coatings [4]. Following, copper and other metals were used as alternatives to the paints. But due to its high persistence and toxicity, these compounds posed unfavorable medium for the aquatic life and ocean environment [5]. Especially,

after the tributyltin (TBT) was banned all over the world, researchers have been looking into the environmentally-friendly coatings to replace them. Among that, two effective strategies have been established to deal with these nonspecific protein adsorption and bacterial adhesion. The first is to use contact killing mechanism to kill effectively the microorganism attached [6,7]. The second is to use non-fouling materials to prevent the adhesion of protein and bacteria [8].

Furthermore, as the researchers had ability to understand the anti-fouling mechanism deeply, they began to pay attention to the surface modification for better anti-biofouling purpose. In view of the surface chemistry, many studies have found that one promising strategy to avoid the adhesion of microorganism is grafting functionalized polymer brush, because the corresponding modified surfaces are more sufficient to prevent the bio-fouling, which are attributed to the high grafting density and molecular weight of polymer brushes [9,10]. Experimental results also indicated that the amphiphilic copolymers, zwitterionic polymers and hydrophilic polymer chains are more suitable for the

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marine antifouling [11–15]. Especially, the zwitterionic polymers (phosphobetaine, sulfobetaine and, carboxybetaine, etc.) with a mixture of anionic and cationic terminal groups possess good antifouling properties, since they can resist nonspecific protein adsorption via a bounded hydration layer induced by the solvation of charged terminal groups, in addition to hydrogen bonding interactions [16]. Meanwhile, they show greater potential in the field of biomedicine as alternatives to PEGylation.

More importantly, zwitterionic components are easy to be functionalized with broader chemical diversity and greater freedom for molecular design [17–20]. Among that, sulfobetaine methacrylate (SBMA) is more promising for a wide range of applications due to its biometric nature, ultra-low fouling properties, stability, and the commercial availability of the monomer [16,21,22]. For example, the Bailey and workers prepared PSBMA brushes via SI-ATRP onto the silicon surface through precursor silane SAMs of undecyltrichlorosilane *a*-bromoisobutyrate. In comparison to unmodified surfaces, they found the modified surfaces have better antifouling properties against undiluted FBS [23]. Chang et al. researched various statistical poly(SBMA-co-NI-PAAm) copolymers, and the amount of adsorbed plasma proteins was found to be extremely low [24].

On the other hand, different from the chemical modification by surface coating, bio-inspired structure has been occupied with effectively self-cleaning and anti-fouling as well, such as lotus leaf, shark-skin, ice leaves and butterfly wings in living nature [25], because these specialized surface topographies could largely impede the algae settlement or adhesion behavior. For example, compared with the smooth silicone elastomer, the anion polymer brush grafted silicon replicas prepared by imitating the structure of leaves *Trifolium* could effectively avoided the adhesion of microalgae (*Chlorella* and *Nannochloropsis*) [26]. Furthermore, effective fouling release properties were exhibited for the hydrophilic 3-sulfopropyl methacrylate potassium salt (SPMA) polymer brush modified structured surface. In addition, it is also believed that the combination of surface chemistry and structure provides a promising strategy for the preparation of environmentally friendly anti-biofouling materials [25]. Whereas, as it has been widely acknowledged that the anionic components were subject to the ion shielding effect inevitably, and the corresponding hydration effect would be whittled accordingly upon salt ion medium.

Based on previous work [27], a one-step (SSI-ATRP) method was designed and employed to construct a novel surface on which the surface chemistry and topography can be induced at once. Interestingly, the synergistic antifouling performance of composition and structure, which was prepared by grafting SPMA on the flat hydrophilic resin surface using SSI-ATRP, was achieved. More importantly, considering that the interface hydration effect, ion of the anionic SPMA polymer brushes modified resin surface would be whittled when they are faced with high concentrations of salt ions, thereby weakening the corresponding antifouling capacity, while zwitterionic SBMA component with persistent stability is almost not affected by the salt ions, SBMA components was, therefore, introduced into the anionic SPMA system to

improve the overall resistance to biological fouling of single SPMA system (Scheme 1). Experimental results have shown that the developed SSI-ATRP strategy could endow general surface with intelligent surface chemistry and topography concurrently as well as improved antifouling behavior, and the durability of synergistic antifouling performance to external ion environment also can be greatly improved by integration of zwitterionic zwitterionic. This novel surface construction technology shows divers applications in the field of the marine antifouling and some other relevant aspects.

2. Experimental section

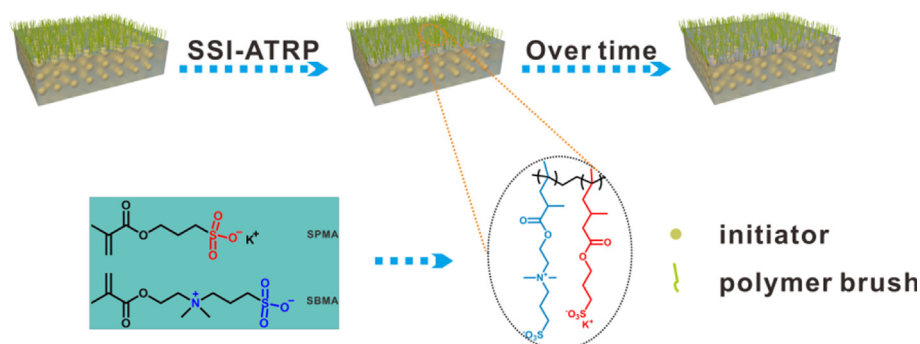
2.1. Materials

The following materials were used in this work: butyl methacrylate (BMA, 98%), methyl methacrylate (MMA, 99%), butyl acrylate (BA, 98%), 1-vinyl-2-pyrrolidone (NVP, 99%), 2-hydroxyethyl methacrylate (HEMA, 97%), toluene (99.5%), 2,2-azobisisobutyronitrile (AIBN), dichloromethane (DCM, 99.5%), triethylamine (99%), 1,6-diisocyanatohexane (HDI, 99%), copper (I) bromide (CuBr, 99.999%), 2,2'-bipyridine (bipy, 99%), methanol (99.5%). 2-Bromoisobutyryl bromide (BrIBuBr, 98%), 3-Sulfopropyl Methacrylate Potassium salt (SPMA, > 95%) and [2-(Methacryloyloxy)ethyl] dimethyl-(3-sulfopropyl) ammonium hydroxide (SBMA, 97%) were purchased from J&K Chemical Ltd. Both the *Porphyridium* sp. and diatoms *Navicula* sp. (N. sp.) were supported by Freshwater Algae Culture Collection at the Institute of Hydrobiology. All reagents were used as-received, and deionized water was used in all experiments.

2.2. Preparation of resin

The initiator 2-(2-Bromoisobutyryloxy) ethyl methacrylate (Br-MA) was synthesized referencing published procedures [28]. And the chemical structure of Br-MA initiator is shown in the Supporting Information (Scheme S1). The initiator chemically embedded resin (i-resin) with hydrophilic property was prepared by a feasible radical copolymerization. Here, a mixed solution containing BA (1 g) and toluene (20 g) in a flame dried round-bottom flask was heated to 80 °C under argon atmosphere. And then a mixture containing BMA (7.00 g), MMA (3 g), BA (1 g), NVP (2 g), HEMA (4 g), Br-MA (2 g), AIBN (0.4 g), toluene (5 g) was added dropwise. The solution was stirred and the radical polymerization was expected to take place within 4 h, then adding a number of AIBN and toluene (5 g) into the mentioned reactor. The polymerization was completed after another 2 h reaction. Before the solution was spun on the dried glass substrates with about 100 μm thicknesses, a fixed amount of curing agent 1,6-diisocyanate was added and degassed for 10 min. At last, the sample was placed in the oven at 80 °C for 5 h to ensure that the chemical cross-linking was occurred absolutely.

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.eurpolymj.2018.07.025>.



Scheme 1. Diagram of the SSI-ATRP process on initiator-embedded hydrophilic resin substrates. The initiating groups are covalently embedded in the substrate, with the increase of polymerization time, the sub-surface of initiator embedded into substrates swelled and triggered ATRP in situ to form thick and robust structured polymer brush interface.

2.3. Preparation of polymer brush modified structured substrate

SPMA, SBMA polymer and SBMA-co-SPMA copolymer brushes tethered structured substrates were prepared by a novel method, sub-surface initiated atom transfer radical polymerization (SSI-ATRP). It is a facile method, which could modify the surface with multi-functions permanently, such as the common resin and other materials. Namely, an allyl-terminated initiator (Br-MA initiator, in Scheme 1) was copolymerized with the acrylic monomer to prepare the initiator embedded acrylic resin (i-resin), and the corresponding acrylic coating can be fabricated via curing agent HDI. The experimental results showed that the thickness of the polymer brush obtained by SSI-ATRP method was at the micron level and well defined surface topography was induced concurrently. The developed surface modification strategy would have a good advantage in antifouling since either surface chemistry or morphology can be easily regulated. Because the ATRP process occurred on the range from the surface to sub-surface (generally micro level), polymer brush prepared through SSI-ATRP was capable of better stability and durability than those polymer brushes prepared using common SI-ATRP method. Moreover, the general ATRP polymerization procedure also was applicable to the SSI-ATRP as follows, firstly dissolving the monomer in water or water/methanol mixed solvent at room temperature and degassing with dry N_2 for 30 min under stirring condition. Next, bipy and CuBr were added to this solution successively. The mixture was then further stirred and degassed with a stream of dry N_2 until a clear solution formed. In the final, the mixture was syringed into a schlenk flask, where i-resin coated substrate was sealed with N_2 protection. After polymerization for a certain time, the samples were taken out, washed with water and ethanol, and dried under N_2 flow for further characterizations. Polymerization recipes for the three monomers are as follows: SPMA 6 g, bipy 80 mg, CuBr 35 mg, 12 mL of water/methanol (v/v = 1:1); SBMA 4 g, bipy 120 mg, CuBr 40 mg, and 8 mL of water/methanol (v/v = 1:1). SPMA-SBMA (1:1) 6 g, bipy 120 mg, CuBr 40 mg, 12 mL of water/methanol (v/v = 1:1).

3. Results and discussion

3.1. Characterization of the samples

At the beginning, the mechanical property of the resin was characterized. The tensile test results showed that the stress of i-resin-10% (10% represents the curing rate, namely, 10% molar fraction of HEMA was reacted with HDI) could reach 3.8 MPa and the corresponding ruptured strain was 168% (in Fig. S2), which indicated the stability of the resin was favorable. Following, the gel permeation chromatography (GPC graph showed in Fig. S3) and TG-DSC (in Fig. S4) measurements stated the higher molecular weight (1.7×10^4 – 3.4×10^5) and suitable glass transition temperature ($\sim 16^\circ\text{C}$), which accounted for the excellent adhesion strength of the resin itself. Considering that the

average surface temperature of the ocean is about 18°C and the resin, therefore, would not change with the external stress in the marine environment, so that it can be more durable with the hull.

Afterwards, the chemistry characterization of SPMA, SBMA, and copolymer brush modified resin, was assessed using the attenuated total reflectance-Fourier transform infrared spectroscopy (ATR-FTIR, Nicolet iS10, Thermo Scientific, USA) with a resolution of 4 cm^{-1} . The ATR-FTIR spectrums are shown in Fig. 1a. Compared with the blank sample, the SPMA and SBMA polymer brush grafted resin showed obvious characteristic peaks of essential functional groups, which proved that the anionic and zwitterionic polymer components were polymerized and tethered successfully to the resin through SSI-ATRP. To further demonstrate polymerization process, the surface chemical composition was characterized by XPS (as shown in Fig. 1b) using a Thermo ESCALAB 250Xi spectrometer with a monochromatic Al K α radiation, and the binding energy was referenced to the C 1s line at 284.6 eV from adventitious carbon. A distinct Br 3d peak at 70 eV existed in blank sample (i-resin), which indicated that the initiator was successfully mixed into the substrate. It also proved that SPMA, SBMA polymer brushes have been successfully polymerized through SSI-ATRP because an obvious S 2p peak at 168 eV could be observed and Br 3d peak at 70 eV disappeared for samples. Similarly, S 2p peak at 168 eV and N 1s peak at 400 eV appeared and Br 3d peak at 70 eV disappeared for the copolymer brush modified resin, which indicated the copolymers successfully grafted to the sample.

In addition, hanging fluid shear experiment simulating the ocean wave flow condition was carried, which showed that resin was still intact after three months immersion (Fig. S5). This result indicated that the binding force of the resin in metal plate was fine. At the same time, the durability of the SPMA polymer brushes prepared by SSI-ATRP and SI-ATRP methods respectively was compared in fluid shear experiment simulating the ocean wave flow condition. As shown in Figs. S5 and S6, the surface of sample prepared from SSI-ATRP was still super-hydrophilic ($\text{CA} < 10^\circ$) after three months immersion. In marked contrast, the contact angle of the sample prepared from SI-ATRP increased significantly range from 2.2° to 52.3° after one month immersion. To sum up, compared with the SI-ATRP, the polymer brush polymerized on the resin surface by SSI-ATRP was very durable under seawater.

Static contact angle (CA) values were acquired at an optical contact angle meter (DSA100, Krüss Optronic GmbH, Deutschland) at ambient temperature (25°C). A droplet of $5\text{ }\mu\text{L}$ deionized water was dropped on the samples using an automatically dispense controller, and the contact angles were determined automatically using the Young-Laplace fitting algorithm. The average CA values were obtained by measuring the sample at three different positions of the substrate. The results are shown in Fig. S7. It was obvious to observe that the contact angle was less than 10° for the SPMA brush coated surface, which indicated SPMA polymer brush is hydrophilic and a thick layer of hydration will be

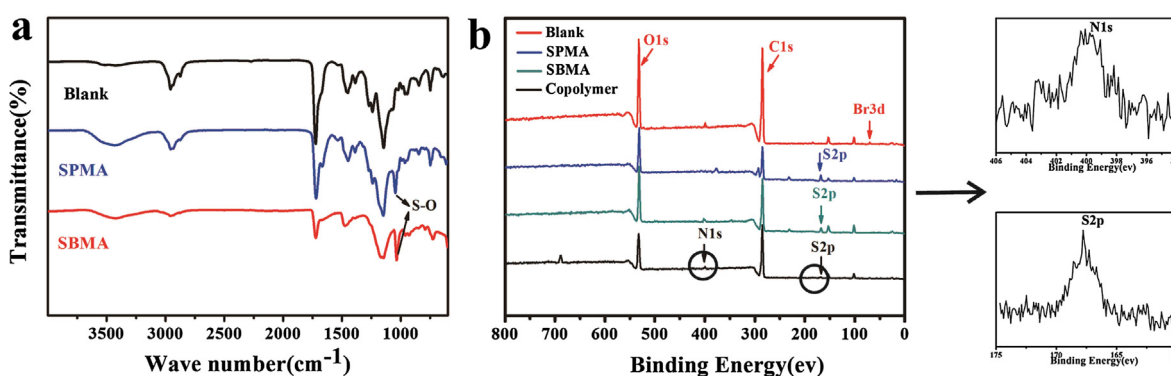


Fig. 1. (a) The ATR-FTIR spectrums of SPMA, SBMA polymer brushes modified substrates. (b) XPS characterization confirmed successful graft of SPMA, SBMA and copolymer brushes. All of the spectrums were aligned C1s peak.

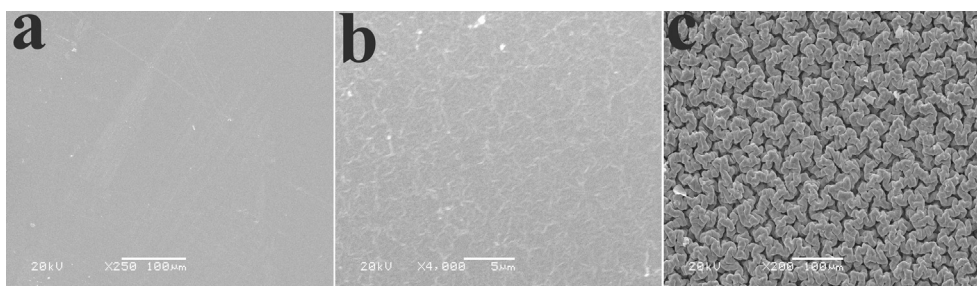


Fig. 2. The surface topographies of samples, (a) blank, (b) grafting SBMA polymer by SI-ATRP, (c) by SSI-ATRP.

formed in water. Similarly, for the substrate of the polymerized SBMA brush, the contact angle reached 20° , which also showed the hydrophilic state. On the other hand, the effect of salt ion on the hydration behavior of various polymer brushes was also investigated through contact angle characterization (Fig. S8). It was clear that both SPMA (a1) and SBMA (a2) polymer samples have significant changes in contact Angle after dealing with Ca^{2+} salt solution.

3.2. Characterization of surface morphology

Inspired by the exquisite micro-structure, we studied the surface morphology induced by SSI-ATRP strategy. Among that, the scanning electron microscope (SEM) images of the blank (Fig. 2a), sample grafting SBMA polymer by SI-ATRP (Fig. 2b) and by SSI-ATRP (Fig. 2c) were shown in Fig. 2. The common SI-ATRP initiator only presented on the substrate surface through chemical vapor deposition (CVD), the polymerization process occurred solely at the interface. Besides, considering the inevitable chain termination reaction mainly induced by impurities, the propagation step would be hampered unfortunately. The thickness of polymer brush prepared using SI-ATRP process for SPMA and SBMA were only about dozens of nanometers in the end. Meanwhile, the thickness of SBMA polymer brush was lower than SPMA ones, which was mainly determined by the variety of monomer reactive activity. However, for SSI-ATRP, the initiator was mixed in matrix using a simple method to form chemical network in resin body. During the process of ATRP, due to swelling effect of the resin in the solvent, polymerizable monomer could constantly reacted with the initiator from resin surface to body as time goes on, making the polymer molecular weight increases greatly. Therefore, compared to the sample prepared by SI-ATRP, it was clear that the surface roughness increased for the SSI-ATRP prepared samples, micron-sized wrinkle can be observed obviously.

At the same time, the dynamic state of the polymer brushes was also

studied, respectively. In reported works, the effects of different parameters such as surface density of bromine, solvent polarity, ligand type, and catalyst-to-deactivator ratio on the kinetics of surface-initiated polymerization have been thoroughly studied [29]. For example, Jiang and coworkers studied the film thickness of SBMA polymer brushes via SI-ATRP as a function of polymerization time. In the initial stage of reaction before 7 h, the film thickness increased linearly with time. After 7 h, the increase in thickness became slower [30]. As depicted in Fig. 3 and Fig. S9, there was no doubt that the thickness of SBMA prepared using SSI-ATRP increased with increasing polymerization time, which is in good agreement with the results reported in the literature [30].

For SPMA polymer brush, the thickness ranged from $50\text{ }\mu\text{m}$ to $97\text{ }\mu\text{m}$ varied with time ranged from 1 h to 2.5 h. And for SBMA polymer brush, the thickness ranged from $7.5\text{ }\mu\text{m}$, $12.5\text{ }\mu\text{m}$, $20\text{ }\mu\text{m}$ to $37.5\text{ }\mu\text{m}$ varied respectively with time ranged from 5 h, 6 h, 7 h and 9 h (Fig. 4). In addition, the surface morphologies and roughness of polymer brushes tethered resin with increasing the polymerization time, which were mainly due to the swelling effect of acrylic resin. Thus, the structure and composition were introduced simultaneously on the substrate surface by one-step SSI-ATRP method and the thickness of the polymer brush has been achieved at micron level.

3.3. Static antifouling properties of samples

The antifouling performance of blank resin, SPMA polymer brush modified surface prepared using SI-ATRP and polymer brush modified structured substrate prepared through SSI-ATRP for the resistance to red algae *Porphyridium* sp. were conducted in the laboratory. As shown in Fig. 5. Meanwhile, the antifouling results of the three samples for the settlement of diatoms *Navicula* sp. showed in Fig. S10. As is seen, after culturing under quasi-static condition for 24 h, there were few amount of algae adhered onto the SPMA polymer brush modified structured

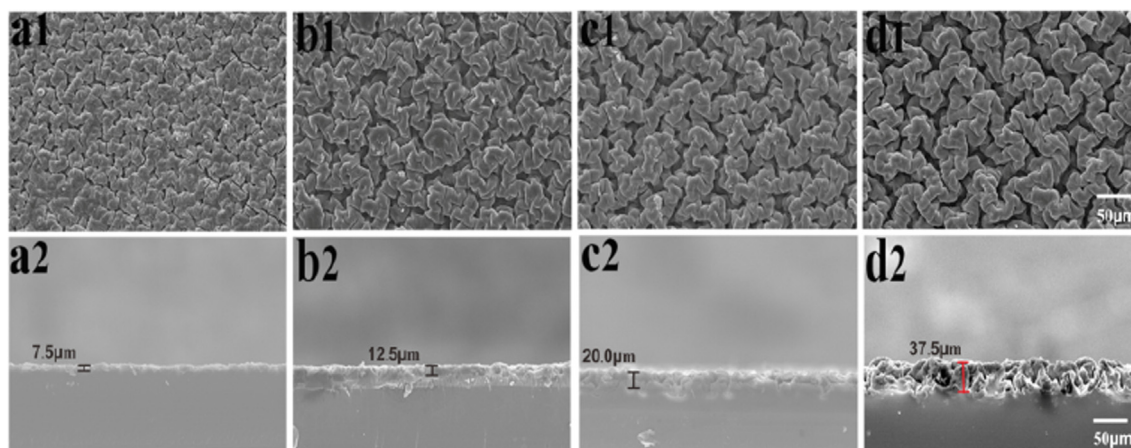


Fig. 3. The surface topographies and thickness of the SBMA polymer brush modified hydrophilic resin substrate vary with time range from 5 h, 6 h, 7 h, and 9 h in same condition.

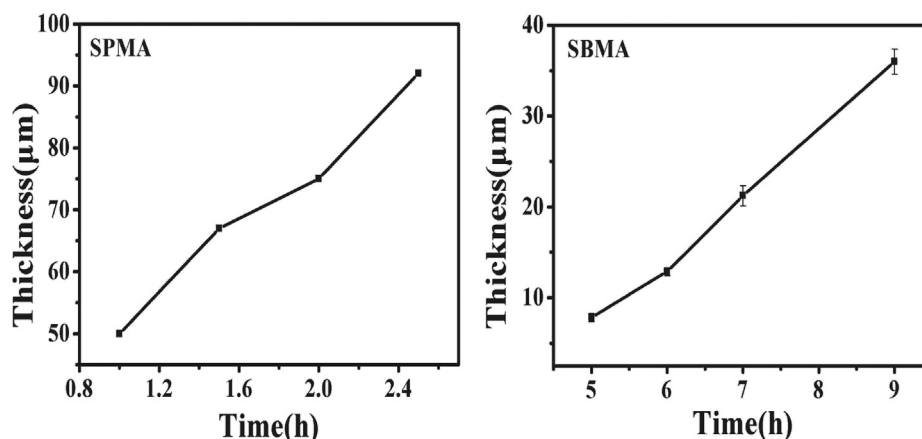


Fig. 4. The data of polymer brush thickness changes over time. The thickness of the two polymer brushes distinctly increased over time.

surface compared with control resin and unstructured surface. And the corresponding fouling densities were 2206, 145, 26/mm² on the three samples for red algae *Porphyridium* sp., respectively. In the same way, the fouling densities were 2391, 88, 19/mm² on the three samples for the diatoms *Navicula* sp., respectively. Interestingly, these experimental results prove that the combination of structure and chemical composition has ideal synergistic antifouling effect.

However, because the anionic polymer brushes are salt-sensitive, there are more amount of algae adhered onto the SPMA polymer brush modified substrate after they are pre-soaked in 1 M Ca²⁺ for five hours. The fouling densities were increased up to 429, 83/mm² for red algae *Porphyridium* sp. (Fig. 6a) and diatoms *Navicula* sp. (Fig. S11a). Consistent with the contact angle characterization, in which the corresponding contact angle changed from 5.4° to 72.2° after soaking in salt solution for five hours, a deteriorated anti-fouling effect for SPMA component were exhibited after soaking in salt solution. It was mainly attributed to the decreased hydration effect induced by the salt ion screen effect. At the same time, the effect of salt ion on the antifouling performance of zwitterionic ionic liquid SBMA polymer brush was also studied before and after the regulation of salt ions medium. As shown in Fig. 6b, different from bio-fouling behavior of SPMA polymer brush modified surface, the red algae *Porphyridium* sp. fouling densities for the SBMA polymer before and after treated with salt ion were 144 and 168/mm², respectively. Besides, a similar bio-fouling behavior also can be observed for the settlement of diatoms *Navicula* sp. (Fig. S11b), and the corresponding fouling densities were 159 and 248/mm², respectively. These results illustrated that the antifouling effect of zwitterionic ionic liquid SBMA polymer brush was almost unchanged, and ideal anti-fouling effect was still maintained even for the high salt ion environment treated SBMA polymer. As a consequence, the zwitterionic ionic liquid not only has prior resistance to the adhesion of micro-organisms, but also occupies robust endurance to the salt ion treatment, which might be due to the hydrophilic characteristic and charge stability of the zwitterionic SBMA polymer compared with other hydrophilic materials [31].

Considering the antifouling behavior of anionic SPMA component and its salt ion sensitivity, the antifouling effect of SPMA-co-SBMA copolymer modified structured surface was further studied. The result of the anti-fouling for the red algae *Porphyridium* sp. is shown in Fig. 7. The fouling densities were 98 and 108/mm² on samples respectively for the copolymer before and after treated with salt ion. In addition, the result of the anti-fouling for diatoms *Navicula* sp. shown in Fig. S11c and the fouling densities were 111 and 100/mm² on samples before and after treated with salt ion, respectively. It was easy to find that the salt ion responsiveness of anionic polymer can be effectively shielded by integrating with SBMA polymer, and a persistent antifouling behavior can be obtained before and after treated with salt ion medium. The experimental result indicated that by copolymerization with zwitterionic polymer component, the challenge of worse anti-fouling effect caused by decrease hydration for SPMA polymer can be overcome. Correspondingly, it can achieve stable and ideal anti-fouling effect under seawater regardless of salt ion circumstances.

3.4. Ocean field assay of polymer brushes samples

To demonstrate the practicability of the as-prepared samples, blank substrate and samples which were SPMA and SBMA polymer brushes modified were held to the steel plates for an ocean field assay. The field immersion tests were carried out in natural seawater from December 2017 to January 2018 (Nan'ao Sea, Shenzhen, +23°28'N, +117°35'E) for 45 days. The samples mounted onto stainless steel plate supports were immersed into seawater at a depth of about 1 m. Fig. 8 shows the photographs of bio-fouling assessment of samples.

For the second week immersion, all samples, including the control plate (resin on steel plates), showed barely any sign of bio-fouling but some sea mud. After four weeks, the control plate began to be fouled by barnacle larva, which might mean the formation of “condition film” on the control surface [9,32], while there was no obvious bio-fouling phenomenon appeared for both SPMA and SBMA embedded structured substrates except sea mud. For six weeks of immersion, serious bio-

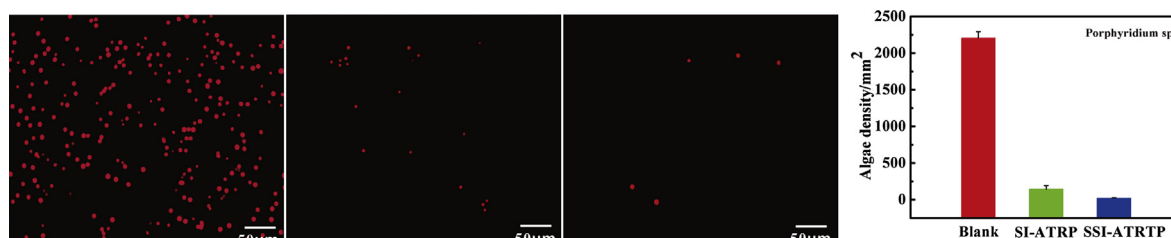


Fig. 5. The algae cell density of red algae *Porphyridium* sp. on the blank substrate, SPMA polymer brush modified unstructured surface and SPMA polymer brush modified structured surface.

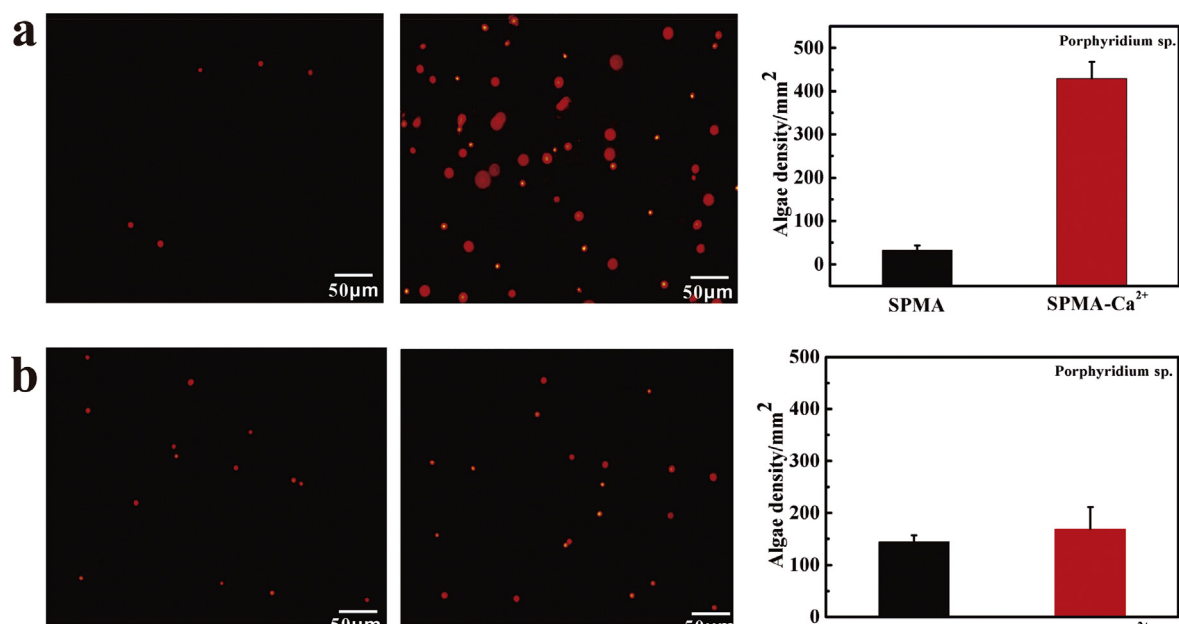


Fig. 6. (a) The algae cell density of red algae *Porphyridium* sp. on the SPMA polymer brush modified structured surface before and after ion regulation respectively. (b) The algae cell density of red algae *Porphyridium* sp. on the SBMA polymer brush modified structured surface before and after ion regulation respectively.

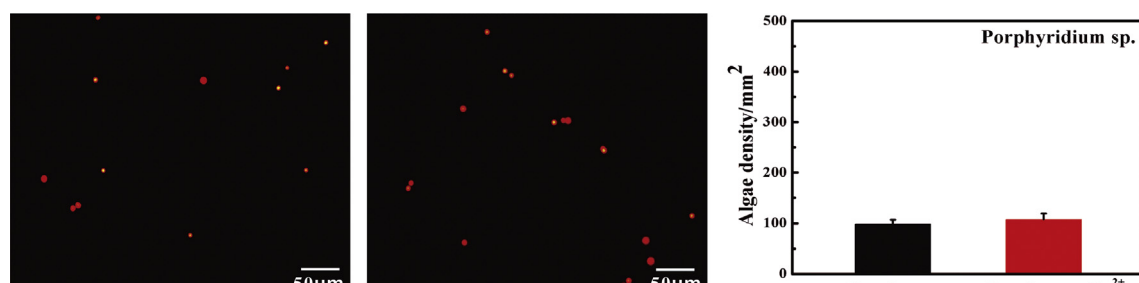


Fig. 7. The algae cell density of red algae *Porphyridium* sp. on the SPMA-co-SBMA copolymer brush modified structured surface before and after ion regulation respectively.

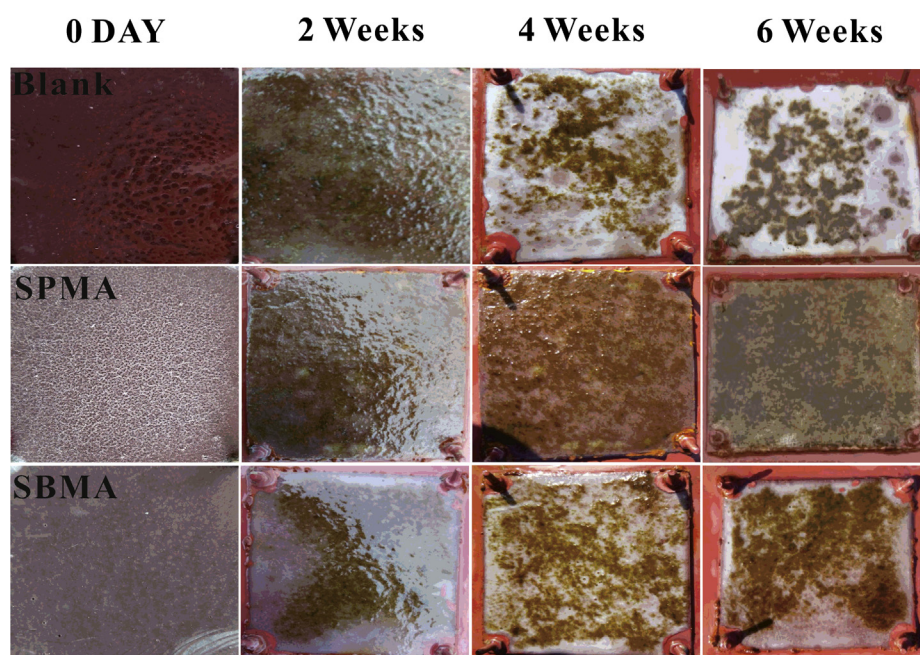


Fig. 8. The photographs of bio-fouling assessment of samples for an ocean field assay.

fouling phenomenon can be observed for the control surface and the bio-fouling was mainly dominated by barnacle larva and *Callysponginvaginalis* larva. However, the performance of as-prepared structured polymer surfaces were different from the control one, either SPMA or SBMA integrated resin still remained relatively clean and there was only very small amount of *Callysponginvaginalis* larva attached on the samples, which meant that the samples combining surface structure with chemical composition possessed superior antifouling effect and mechanical stabilities under seawater. These experimental results are accorded with the static antifouling experiment. Overall, despite the unexpected but negligible damages on the as-prepared samples, such as some small cuts and wear, the active components integrated with structure character truly provides an excellent resistance to micro-organism accumulation for long-term application. Besides, the polymer brush-based structured materials are beneficial to environmental protection, could deliver multiple green-economy benefits, and exhibit the potential application in the field of marine bio-fouling.

As a conclusion, a novel methodology of SSI-ATRP was proposed to avoid the problem that the polymer brushes would fall off in the short term. Compared with the method of traditional SI-ATRP, the thickness of polymer brushes greatly increases from nano-scale to micro scale, and the enough interface adhesion strength of polymer brushed modified coating to various substrate is guaranteed by the acrylic resin which could firmly held itself to the steel plate for ocean field application. More importantly, the proposed method mentioned above could prepare the chemical modified but structured substrate by one-step polymerization process and make the substrate more stable for a long-term practical application. In the long run, an appropriate collaborative design combined structure with chemical component by the technology of SSI-ATRP for the marine anti-biofouling should be developed in the future.

4. Conclusion

In present work, the novel sub-surface atom transfer radical polymerization (SSI-ATRP) was proposed, which could introduced surface chemical composition and specialized structure in the resin base at the same time. And we studied synergistic marine anti-biofouling effect by one-step method. Compared with surface atom transfer radical polymerization (SI-ATRP), it gave the coating and polymer brushes better stability and longer term antifouling properties. The antifouling performances of SPMA and SBMA polymers modified structured surfaces were studied, respectively. The experimental results indicated that the anionic SPMA polymer brush modified surface occupied excellent antifouling properties. However, their further application was hampered by salt ion shielding effect. To overcome this problem, zwitterionic integrated SBMA-co-SPMA copolymer brushes modified structured surfaces were fabricated, which could improve the salt ion responsiveness of anionic SPMA by taking advantages of the stability of zwitterionic polymer. More importantly, the copolymer might be an ideal antifouling candidate with stable and persistent hydration effect. This research has certain value to marine antifouling in the long run.

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